Exercise 2.8

Given

$$S_0(x) = e^{-0.001x^2}, \qquad x \geq 0,$$

find expressions for (a) and (b), simplifying as far as possible,

(a) $f_0(x)$, and

(b) μ_x.

Solution

(a)

Since

$$S_0(x) = 1 - F_0(x),$$

we have

$$F_0(x) = 1 - S_0(x) = 1 - e^{-0.001x^2}.$$

The density function is

$$f_0(x) = \frac{d}{dx} F_0(x).$$

Thus,

$$\begin{aligned} f_0(x) &= \frac{d}{dx}\left(1 - e^{-0.001x^2}\right) \\ &= -e^{-0.001x^2}(-0.002x) \\ &= 0.002xe^{-0.001x^2}. \end{aligned}$$

Therefore,

$$\boxed{f_0(x) = 0.002xe^{-0.001x^2}}.$$

(b)

The force of mortality is

$$\mu_x = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x).$$

Now,

$$\begin{aligned} \frac{d}{dx} S_0(x) &= \frac{d}{dx} e^{-0.001x^2} \\ &= e^{-0.001x^2}(-0.002x). \end{aligned}$$

Therefore,

$$\begin{aligned}\mu_x &= -\frac{1}{e^{-0.001x^2}}\left[e^{-0.001x^2}(-0.002x)\right] \\ &= 0.002x.\end{aligned}$$

Thus,

$$\boxed{\mu_x = 0.002x}.$$